

01 DECEMBER

RAG

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What is RAG? Retrieval Augmented Generation

How do you build RAG (External Knowledge Augmentation)?

What is Augmentation? Creating new knowledge

1) Manual Context Injection

In this context ^{relevant info} give me the answer.
manually copy & pasting in prompt.

Explain cloud to Brd grade student in three bullet points.
↓
context.

1)

User Prompt: Analyze this customer feedback.

2) Custom Prompt Template:

Most companies are generating custom templates and asking employees to use that.

TEMPLATE: "You are a {role}. Given this {data-type}, {data}.

- Metadata: {source}
- Date: {Date}
- category: {category}

Please {task} while focusing on {focus-area}

Usage :

" You are a financial Analyst. Given this quarterly report, {Q3 Earning Data}.

Metadata :

- Source : Sec Filing.
- Date : 01 Dec 2025
- Category : Financial.

Please identify risk while focusing on cash trends.

3) WEB SEARCH + MANUAL COPY

↳ Go to Google search for Latest FDA approval guideline 2025.

↓

Copy guidelines into Chatgpt & now we can work based on guidelines

4) FINE TUNNING & Adapter training. (Not Reu

For Example: We want to modify the weight in LLM. We have our own dataset.

A medical institution fine tunes ~~the~~ a model on specific clinical protocols.
How to train our own data..

OWN DATASET

5. WEB SEARCH + LLM.

ASK CHAT GPT.

What is the temp now in Hyderabad?

This searches web (ChatGPT Browse).

if we use API, it won't search web.

(INFORMATION RESOURCE)

6. INTERNAL IR + RAG. VECTOR DB & LLM

Structured KB (DB, JSON, HTML, PDF) ...
(and feed into LLM)

Own Knowledge Base.

PDFs → 10,000 → VECTOR DB ← LLM

① What is our refund policy?

↓
convert into Embedding (Numerical values)

↓
search in Vector (Numerical values).

☐ ☐ ☐

LLM finds 3 sentences (close matches)

Embedding is converting text into numerical values.

1) Agentic RAG / Tool-Augmented Agents

Agentic means LLM ~~can reason~~ an autonomous agent that can reason & use multiple tools until it reaches a decision.

"Analyze our competitors pricing strategy & recommend changes."

1. I need competitor pricing data. → call
- 2) I need current pricing. - call internal
- 3) Check recent market trends - calling web
- 4) Let me calculate price elasticity - use
- 5) Validate against margins. - call financial API
- 6) Synthesize all data - provide comprehensive recommendation

Real world use case

- 1) Plan trip to Malaysia under \$3000
- 2) VS Code has Code Max agent. Debug my api time has spiked.

Normal RAG Vs Agent.

~~by~~
~~agent~~

1 - 3 Human does the work

4) Model is permanently modified.

5) Automated - external web search.

6) Automated - internal search

7) Autonomous

&

Tomorrow simple RAG, HYBRID RAG, AGENTIC RAG, LLM RAG

